



# Solving simple linear equations with a multiplicative step

## Task A: Multiplicative steps using representations

- 1) Match up the bar models with the calculations which will find the solution. There may be more than one calculation for each diagram.

a) 

|     |     |
|-----|-----|
| $x$ | $x$ |
| 4   |     |

 $4 \times \frac{1}{2}$   
 $4 \div 2$

d) 

|                |  |
|----------------|--|
| $\frac{1}{4}x$ |  |
| 4              |  |

 $4 \div \frac{1}{4}$   
 $4 \times 4$

b) 

|     |     |     |     |
|-----|-----|-----|-----|
| $x$ | $x$ | $x$ | $x$ |
| 8   |     |     |     |

 $8 \times \frac{1}{4}$   
 $8 \div 4$

e) 

|                |  |
|----------------|--|
| $\frac{1}{4}x$ |  |
| 8              |  |

 $8 \div \frac{1}{4}$   
 $8 \times 4$

c) 

|                |  |
|----------------|--|
| $\frac{1}{2}x$ |  |
| 4              |  |

 $4 \times 2$   
 $4 \div \frac{1}{2}$

- 2) Use these bar models to find the value of  $x$

a) 

|     |     |     |     |     |
|-----|-----|-----|-----|-----|
| $x$ | $x$ | $x$ | $x$ | $x$ |
| 15  |     |     |     |     |

 $15 \times \frac{1}{5}$   
 $x = 3$

d) 

|     |     |     |
|-----|-----|-----|
| $x$ | $x$ | $x$ |
| 5   |     |     |

 $5 \times \frac{1}{3}$   
 $x = \frac{5}{3}$

b) 

|     |     |     |
|-----|-----|-----|
| 15  |     |     |
| $x$ | $x$ | $x$ |

 $15 \div 3$   
 $x = 5$

e) 

|     |     |     |     |
|-----|-----|-----|-----|
| 18  |     |     |     |
| $x$ | $x$ | $x$ | $x$ |

 $18 \div 4$   
 $x = \frac{9}{2}$

c) 

|     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| 30  |     |     |     |     |     |     |     |     |     |
| $x$ | $x$ | $x$ | $x$ | $x$ | $x$ | $x$ | $x$ | $x$ | $x$ |

 $30 \div 10$   
 $x = 3$

f) 

|                |  |
|----------------|--|
| $\frac{1}{5}x$ |  |
| 10             |  |

 $10 \times 5$   
 $x = 50$

**Task B: Writing solutions algebraically**

1) Solve each equation.

$$\begin{aligned}\text{a)} \quad 2x &= 18 \\ 2x \div (2) &= 18 \div (2) \\ x &= 9\end{aligned}$$

$$\begin{aligned}\text{b)} \quad 6x &= 30 \\ 6x \div (6) &= 30 \div (6) \\ x &= 5\end{aligned}$$

$$\begin{aligned}\text{c)} \quad 5x &= 9 \\ 5x \times \frac{1}{5} &= 9 \times \frac{1}{5} \\ x &= \frac{9}{5}\end{aligned}$$

$$\begin{aligned}\text{d)} \quad 21x &= -21 \\ 21x \div (21) &= -21 \div (21) \\ x &= -1\end{aligned}$$

$$\begin{aligned}\text{e)} \quad -2x &= -12 \\ -2x \div (-2) &= -12 \div (-2) \\ x &= 6\end{aligned}$$

$$\begin{aligned}\text{f)} \quad -8x &= 18 \\ -8x \div (-8) &= 18 \div (-8) \\ x &= -\frac{9}{4}\end{aligned}$$

$$\begin{aligned}\text{g)} \quad \frac{1}{4}x &= 13 \\ \frac{1}{4}x \times (4) &= 13 \times (4) \\ x &= 52\end{aligned}$$

$$\begin{aligned}\text{h)} \quad \frac{2}{3}x &= -10 \\ \frac{2}{3}x \times \left(\frac{3}{2}\right) &= -10 \times \left(\frac{3}{2}\right) \\ x &= -15\end{aligned}$$

$$\begin{aligned}\text{i)} \quad -1.4x &= 4.2 \\ -1.4x \div (1.4) &= 4.2 \div (1.4) \\ -x &= 42 \div (14) \\ -x &= 3 \\ x &= -3\end{aligned}$$

